

Effects of Transcranial Direct-Current Stimulation on Hand Redirection

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ABSTRACT

In virtual reality (VR), hand redirection (HR) has been proposed to enhance the flexibility of haptic feedback. HR takes advantage of the dominance of visual signals over proprioception signals in hand movements, which is achieved by shifting the posture and position of the virtual hand from its physical posture and position. The user does not notice this manipulation within a certain range, but the discomfort becomes apparent when the manipulation is applied beyond a certain range. Given that attention plays a pivotal role in the sensory integration of vision and proprioception, we hypothesized that interventions targeting the left dorsolateral prefrontal cortex (DLPFC), a region associated with attention, could modulate the effects of HR. This study assessed HR detection thresholds (DTs) during left DLPFC intervention using transcranial direct current stimulation (tDCS). Anodal stimulation was applied to enhance left DLPFC activity, and DTs were measured before and after the stimulation. A sham stimulation condition served as a control. In both stimulation conditions, we systematically varied the magnitude of HR from -20 deg (leftward shift) to 20 deg (rightward shift). Based on a user study, our findings revealed that DTs for leftward shifts significantly increased in the sham condition, likely due to habituation. Interestingly, tDCS effectively mitigated this increase in DTs. These results suggest that tDCS may reset the effects of HR habituation or enhance attention for HR.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Recently, an approach using illusions to provide appropriate haptic feedback for virtual objects has been proposed. One common and powerful type of illusion for interaction in virtual reality (VR) is hand redirection (HR), which is achieved by exploiting the dominance of visual over proprioceptive stimuli in the perception of hand movement. HR displaces the visual and actual hand positions and orientations. This manipulation allows the user to obtain haptic feedback of virtual objects with different positions, postures, shapes, weights, and inertia, even though the user is actually touching a single physical proxy. However, it is important to note that HR cannot be extended indefinitely. Users will become aware of the intervention when the discrepancy between visual and proprioceptive signals becomes too significant.

Meanwhile, based on past neuroscience findings, it can influence human sensory-motor systems by intervening in brain activity [4]. Neuroscientific literature suggests that the dorsolateral prefrontal cortex (DLPFC) directs attention toward internal representations of sensory stimuli and motor plans stored in more posterior regions [1]. Therefore, we thought that it might be possible to alter

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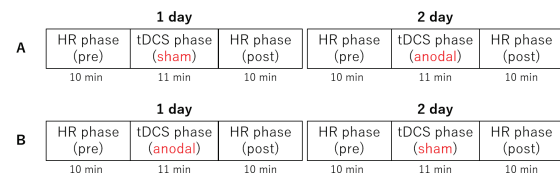


Figure 1: Experimental Procedure. The experiment consisted of an HR phase and a tDCS phase, respectively. The order of the tDCS conditions (sham first for A and anodal first for B) was counterbalanced among participants.

the effects of HR by intervening in the DLPFC. One method of modulating brain activity is transcranial direct current stimulation (tDCS). tDCS is a type of neuromodulation in which a constant, low-level direct current is delivered through electrodes placed on the head. A study has reported that anodal tDCS applied to the DLPFC enhances cognitive control over attentional bias-related items that are behaviourally irrelevant, allowing for their more efficient disregard [3]. Furthermore, numerous previous studies have reported an association between tDCS and hand movements [4]. Based on the above, this study investigates the changes in HR detection threshold (DT) when anodal-tDCS is applied to the left DLPFC.

2 METHODS

2.1 Participants

Twenty healthy participants were recruited for the study. One participant's data was excluded from the analysis because one could not perform the HR task properly. Here, we report the data from 19 participants (9 females, mean age: 22.4 years, SD: 1.7, range: 19-25, 1 left-handed). After explaining the experiment, written informed consent was obtained. The local ethics committee approved the experimental procedure.

2.2 Procedure

Each participant engaged in experimental sessions spanning two days, as illustrated in Fig. 1. Each day of the experiment consisted of a tDCS phase in which anodal tDCS or sham stimulation was applied and an HR phase in which reaching tasks were performed. The HR phase was included before and after the tDCS phase so that we could measure the effect of tDCS on HR. The experiment was a within-participant design for two tDCS stimulus types (sham and anodal stimulation), and the order was counterbalanced among participants and performed in a double-blind. To minimize the carry-over effects of tDCS, there were at least 48 hours between each tDCS phase. Each participant practiced the HR task several times at the beginning of the first day.

2.3 Hand Redirection

This phase was performed in an experimental environment created in Unity (2019.4.33f1), and participants observed the environment using Meta Quest 2. The participant performed a reaching motion by touching the sphere in front with the index finger of the virtual

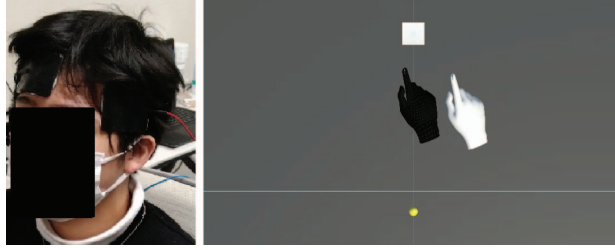


Figure 2: Experimental settings. (Left) A participant was undergoing tDCS. Electrodes were placed on the left temple and right forehead. (Right) Right hand in reaching motion, viewed from above. The white hand is the virtual hand. The black hand is the real hand, invisible to the user, but drawn here for illustration. Participants touched the sphere on the front side (bottom of the figure) with the index finger and then reached the cube on the back side (top of the figure).

right hand and then touching the cube in the back. The virtual hand was redirected from the real hand position measured by the hand tracking of the HMD (Fig. 2 right). Redirection was performed by the hand rotation horizontally by α degrees. In this experiment, α was set to -20, -16, -12, -8, -4, -4, 0, 4, 8, 12, 16, and 20, with ten trials of each gain, for a total of 110 trials tested. The order of α was randomized.

2.4 Transcranial Direct Current Stimulation

Transcranial direct current stimulation was administered via a stimulator (neuroConn DC-Stimulator Plus, neuroCare Group GmbH, Munich, Germany), using silicon rubber electrodes (5 cm $\times$ 7 cm). The stimulation waveform was synthesized in MATLAB 2022a and conveyed to the tDCS apparatus through a digital/analog converter. In the anodal condition, current strength was 1 mA, and duration was 10 min, so the tDCS after-effect lasted throughout the post-HR task (Fig. 2 left). The onset and offset of tDCS were for 30 seconds. In the sham condition, the current strength was 0 mA, but there was onset and offset in the same way to mimic the cutaneous sensation in the anodal condition. We confirmed in the interviews that all participants could not discriminate between the anodal and sham stimulation conditions.

2.5 Data Analysis and Statistics

The following analyses were performed to obtain the DT of HR from the 2AFC answers in the Hand Redirection phase. The obtained data were divided into leftward HR ($\alpha \leq 0$), i.e., HR in the medial direction of the body, and rightward HR ($\alpha \geq 0$), i.e., HR in the lateral direction, and a logistic function was applied to each data group. The α that was 50% of the obtained psychometric function was set as the DT. The DT for redirection to the left and right was measured before and after the anodal/sham tDCS. Repeated-measures 2-way ANOVA (pre-post $\times$ tDCS) was performed on the obtained DTs in the left and right conditions, respectively. Multiple comparisons were performed when there was a significant effect.

3 RESULTS

3.1 Detection Threshold

The DTs obtained from the participants are plotted for each condition (Fig. 3). Repeated-measures 2-way ANOVA (pre-post $\times$ tDCS) was performed on these data in the left and right conditions, respectively. In the leftward HR condition, there was no main effect of pre-post ($p = 0.33, F_{(1,18)} = 1.0, \eta_p^2 = 0.002$), no main effect of tDCS ($p = 0.11, F_{(1,18)} = 2.8, \eta_p^2 = 0.010$). But interaction of pre-post $\times$ tDCS was significant ($p < 0.05, F_{(1,18)} = 6.4, \eta_p^2 = 0.006$). Holm-corrected post-hoc analysis (Wilcoxon signed-rank tests) showed

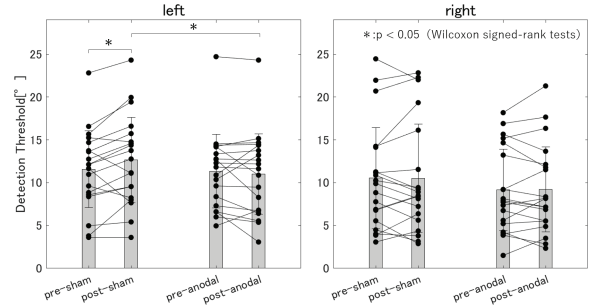


Figure 3: DT for each participant in each condition. The left figure shows the results under the leftward HR conditions, and the right figure shows the results under the rightward HR conditions.

that there was a significant difference between pre-sham and post-sham ($p < 0.05$) and between post-sham and post-anodal ($p < 0.05$) (Fig. 3). In the rightward HR condition, there was no main effect of pre-post ($p = 0.97, F_{(1,18)} = 0.0, \eta_p^2 = 0.000$), no main effect of tDCS ($p = 0.08, F_{(1,18)} = 3.5, \eta_p^2 = 0.014$), and no pre-post $\times$ tDCS interaction ($p = 0.97, F_{(1,18)} = 0.0, \eta_p^2 = 0.000$).

4 DISCUSSION AND CONCLUSION

The observation that DT increased with sham stimulation under the LEFT redirection condition confirms that DT increases with continuous experience of HR. Conversely, there was no significant difference in the leftward DTs before and after tDCS in the anodal condition. In addition, leftward DT after anodal stimulation was significantly smaller than DT after sham stimulation. This suggests that tDCS for DLPFC may have decreased the DT by an amount equivalent to the threshold increase due to habituation and increased attention to HR manipulation. Another interpretation of these results is that adaptation to hand redirection did not occur with tDCS. Previous studies showed that anodal tDCS to the left PFC slowed motor learning [2]. Therefore, anodal tDCS over the left DLPFC may similarly inhibit motor learning.

To ascertain the prevailing mechanism, it is imperative to conduct analyses to determine if motor learning is inhibited, as evidenced by analyzing the duration of reaching motion and motor data. We would also like to study the effect of cathodic tDCS on DLPFC and HR. If it becomes clear that the DT of HR can be expanded by using such brain stimulation methods, it can be expected to lead to engineering applications.

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